



Platform API Audit: Received Surface and Required APIs

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Abstract

Three integration guides have been received from the SF Private Bank platform: Account Management, Virtual Cards, and Crypto Service. They document card-account creation for natural persons, the virtual-card KYC and consent lifecycle, and custodial crypto wallets and swaps. Everything else a complete banking and card program requires — sandbox and production base URLs, user provisioning, identity document submission, individual *and corporate* account opening with IBANs, payment rails, the physical-card lifecycle, and webhooks — is not yet available or not yet documented. This audit records the received surface and the 54 missing or unclear APIs, each as a concrete endpoint contract priced by priority. Section 4 is the detail; the closing request summary is the punch list.

1 Introduction

This document is for the platform integration team. It records what the issued guides provide and the complementary APIs a complete banking and card program still needs — individual and corporate accounts, payments, cards, and crypto, end to end. Where a shorthand helps, the Wise Platform API is the reference shape: each requirement group names its Wise equivalent, meaning “a contract of that familiar form is exactly what we need.”

2 Received Surface

The three integration guides document, in full:

Endpoint	Guide	Purpose
POST /account/:userId	Account Mgmt	Create established card account
GET /account/:userId	Account Mgmt	Account state, KYC state, cards
GET /account/:userId/kyc-status	Account Mgmt	Issuer KYC status
POST /account/:userId/virtual	Virtual Cards	Create virtual-card account
GET /account/:userId/virtual/kyc-url	Virtual Cards	Hosted KYC session URL
GET /account/:userId/virtual/consent-agreement-url	Virtual Cards	Consent agreement URL
(referenced, unspecified) single-card order	Virtual Cards	Order the virtual card

Table 1: Complete issuer API surface received to date.

Endpoint	Guide	Purpose
POST /wallets/custodial/create	Crypto Service	Create custodial wallet (6 tokens)
GET /wallets/custodial/:userId[:token]	Crypto Service	List / read custodial wallets
POST /wallets/custodial/send	Crypto Service	On-chain send (OTP)
POST /otp/send	Crypto Service	OTP for sends and swaps
POST /wallets/custodial/swap[/estimate]	Crypto Service	Custodial swap, estimate/execute (OTP)
POST /swaps/external[/estimate]	Crypto Service	Fiat balance to on-chain token (OTP)
GET /swaps/external/:userId:txId	Crypto Service	External swap status

Table 2: Crypto Service surface received. Production host <https://crypto-v1-api-rdqehkur6a-ey.a.run.app>; the service is production-only today — sandbox calls return an error.

Both card guides assume the user *already exists* upstream and reference an API Reference not yet shared. The account-creation body accepts natural-person fields only — no corporate or business entity form is documented. Neither the base URL nor the authentication scheme has been issued; production credentials do not yet exist. The fee schedule (Platform Commercials) has been received.

3 Gap Summary

Priorities: **P0** blocks launch; **P1** blocks a committed near-term product; **P2** completes the portfolio.

Domain	P0	P1	P2
Platform fundamentals	7	1	—
Identity, IDV & onboarding	5	1	—
Accounts (ewallet + IBAN)	6	2	1
Payments & transfers	6	3	1
Cards (CaaS)	6	6	2
Webhooks & operations	3	3	1
Total: 54	33	16	5

Table 3: Missing issuer APIs by domain and priority.

4 Required APIs

Endpoint names below are proposals in the style of the received guides; equivalent contracts under different names are equally acceptable.

4.1 Platform fundamentals

Wise equivalent: the public versioned API reference — sandbox, token auth, idempotent POSTs, documented errors.

Requirement	Contract	Pri
Base URLs	Sandbox + production hosts for every service — only the Crypto Service production host is issued	P0
Crypto sandbox / testnet	Testnet-backed sandbox for wallets, sends, swaps — the service is production-only today	P0
Authentication scheme	Key issuance, header, rotation, IP allowlist	P0
API Reference / OpenAPI	The reference both guides cite	P0
Error catalog	Codes, retryability, human text	P0
Idempotency	Idempotency-Key on all mutations	P0
Rate limits	Published limits + headers	P0
Environments & fixtures	Sandbox test personas, funding simulator	P1

4.2 Identity, IDV, and onboarding

Wise equivalent: profile creation plus verification document upload, for personal and business profiles, with status webhooks.

Identity is collected on our side — document capture and selfie liveness — and the completed file handed over. The hosted-URL flow works for cardholder KYC; account opening at scale requires server-to-server submission.

Requirement	Contract	Pri
User provisioning	POST /user — create the upstream user both guides presuppose; GET /user/:id	P0
IDV document submission	POST /account/:userId/documents — identity document, proof of address, liveness artifact; multipart + metadata	P0
KYB (business)	Entity documents, UBOs, authorized signers	P0
KYC decision webhook	kyc.approved / kyc.rejected with reason codes	P0
Consent state webhook	consent.accepted	P0
Re-KYC / refresh triggers	Periodic refresh, document expiry events	P1

4.3 Accounts — ewallet and IBAN

Wise equivalent: the multi-currency account and balance APIs — open a balance, read account details (IBAN/BIC), statements.

The fee schedule prices ewallet setup, wallet bank account setup, monthly and annual service, and dormancy. None of these products has an API yet.

Requirement	Contract	Pri
Open ewallet	POST /account/:userId/ewallet	P0
Open IBAN wallet bank account	POST /account/:userId/bank → returns IBAN/BIC — for individuals <i>and</i> corporates	P0
Corporate account opening	Business entity form for account + cardholder creation; the received POST /account/:userId accepts natural-person fields only	P0
Account details	GET .../details — IBAN, BIC, routing, reference	P0
Balances	GET .../balance per currency	P0
Transaction history / statement	GET .../transactions, statement export	P0
Multi-currency sub-balances	USD/EUR/GBP at minimum	P1
Account lifecycle	Close, freeze, dormancy state & fees	P1
Interest / yield products	If offered	P2

4.4 Payments and transfers

Wise equivalent: recipient accounts plus the transfers API — create, fund, track, with delivery estimates and status webhooks.

Every rail on the fee schedule is sold and priced; none has an API. FX and crypto conversion are *not* requested — we execute both ourselves and settle plain fiat over these rails.

Requirement	Contract	Pri
Beneficiary registration	POST / beneficiary + verification state	P0
Domestic ACH (US), in/out	Originate + status lifecycle	P0
Wires — SWIFT, Fedwire	Originate, track, return handling	P0
P2P ewallet transfer	POST / transfer ewallet→ewallet	P0
Ewallet ↔ bank transfer	Internal movement	P0
Payment status webhooks	payment.settled / failed / returned	P0
International ACH (local rails)	Originate + status	P1
Paper checks (US/CA)	Issue + status	P1
Incoming payment notification	Credit webhooks with payer data	P1
Bulk / batch payouts	Batched origination	P2

4.5 Cards — the CaaS surface

Wise equivalent: platform card issuing — order, activate, freeze, PIN, spending controls, real-time authorization webhooks.

The account state exposes `cards[]` with `isLocked`, `isActivated`, `balance`, and a shipped physical card — implying capabilities that have no documented endpoint.

Requirement	Contract	Pri
Single-card order (spec)	The endpoint the Virtual Cards guide references	P0
Physical card order	Plastic + metal, shipping address, status	P0
Card activation	POST / card/:id/activate	P0
Card load / unload	Funding from ewallet or bank balance	P0
Lock / unlock	isLocked is readable; make it writable	P0
Authorization + transaction webhooks	Real-time auth events, clearing, declines	P0
Secure PAN reveal	Tokenized display (SDK / iframe / one-time token)	P1
PIN set / reset	Secure channel	P1
Push provisioning	Apple Pay / Google Pay in-app provisioning payloads	P1
Spending controls	Per-card limits, MCC rules	P1
Replacement / cancel	Lost, stolen, damaged	P1
3-D Secure	Enrollment + challenge callbacks	P1
Disputes / chargebacks	File, evidence, status	P2
Card-to-card transfer	Priced on the fee schedule	P2

4.6 Webhooks and operations

Wise equivalent: webhook subscriptions with signed payloads, plus activity and statement exports.

Requirement	Contract	Pri
Webhook delivery contract	Signed (HMAC), versioned events, retries	P0
Event catalog	Complete typed event list	P0
Webhook replay	Re-fire by event id	P0
Settlement / reconciliation files	Daily program-level settlement report	P1
Fee reports	Itemized fee extraction matching the schedule	P1
Program ledger / float	Program balance and funding API	P1
SLA + status page	Uptime commitments, incident channel	P2

5 Request Summary

Requested, in order:

1. Sandbox *and* production base URLs for every service, the full API Reference the guides cite, the authentication scheme, and sandbox credentials.
2. User provisioning and server-to-server IDV/KYB document submission (§4.2), with KYC decision webhooks.
3. Ewallet and IBAN account opening for individuals *and* corporates, with account details, balances, and history (§4.3).
4. Payment origination on ACH and wires with beneficiary registration and status webhooks (§4.4).
5. The card lifecycle beyond ordering: physical orders, activation, load, lock, and authorization webhooks (§4.5).
6. The webhook delivery contract and event catalog (§4.6); reconciliation and fee reporting to follow.
7. For the Crypto Service: the authentication scheme, and a testnet-backed sandbox — the service is production-only today.

With items 1–3, accounts open — individual and corporate. With 4, money moves. With 5–7, the complete program operates at production quality.